Chapter 4: Preventive Maintenance and Troubleshooting

Instructor Materials

IT Essentials v7.0

Chapter 4: Preventive Maintenance and Troubleshooting

IT Essentials 7.0 Planning Guide

Chapter 4: Preventive Maintenance and Troubleshooting

IT Essentials v7.0

Edited by D. Shaver Fall 2019

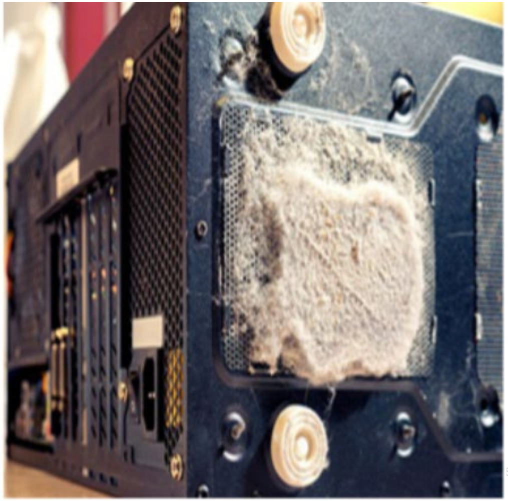
Chapter 4 - Sections & Objectives

- 4.1 Preventive Maintenance
  - Explain why preventive maintenance must be performed on personal computers.
    - Describe PC preventive maintenance.
- 4.2 Troubleshooting Process
  - Troubleshoot problems with PC and Peripheral devices
    - Describe each step of the troubleshooting process.
    - Identify common problems and solutions for PCs.
    - Troubleshoot computer components and peripherals using the six-step troubleshooting process.

4.1 Preventive Maintenance

Preventive Maintenance - Dust

- Clean the computer case exterior
- Dust can travel from exterior to interior through fans
- Dust accumulation
  - Impedes air flow
  - Reduces the cooling of components.
  - Result of Hot computer components?
- Interior dust removal
  - Compressed air
  - low-air-flow ESD vacuum cleaner
  - lint-free cloth



Dust Removal

- When using cleaning products
  - Put liquid on cleaning cloth
  - Wipe outside of the case.

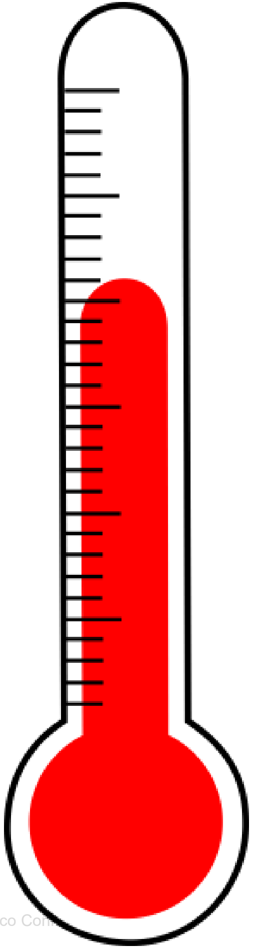
Preventive Maintenance – Internal Components

A basic checklist of components to inspect for dust and damage includes:

- CPU heat sink and fan assembly
- RAM modules
- Storage devices
- Adapter cards
- Cables
- Power devices
- Keyboard and mouse

Preventive Maintenance – Environmental Concerns

- Optimal operating environment
- Clean
- Free of potential contaminants
- Within manufacturer recommended temperature and humidity range



Preventive Maintenance – Software

Verify that installed software is current.

Follow organization policies when installing

- security updates
- operating system
- program updates

Software maintenance schedule:

- Review and install appropriate OS security and driver updates
- Regularly update virus definition files
- Regularly scan for virus and spyware
- Remove unwanted programs
- Regularly scan hard drive for errors

4.2 Troubleshooting Process

Early Troubleshooting

Do you recognize this early computer pioneer?

Introduction to Troubleshooting

- Requires an organized and logical approach to problems.
- Eliminates variables and identifies causes of problems in a systematic order.
- Troubleshooting skills get better with experience.
- Before troubleshooting, protect user data.
- *Ask about floppy disk error*

"Tech support says the problem is located somewhere between the keyboard and my chair."

Troubleshooting Process Steps

Step	Troubleshooting Process
1	**Identify** the Problem
2	Establish a Theory of **Probable Cause**
3	**Test** the **Theory** to Determine Cause
4	**Establish** a **Plan of Action** to Resolve the Problem and Implement the Solution
5	**Verify Full System Functionality** and, if Applicable, Implement Preventive Measures
6	**Document** Findings, Actions, and Outcomes

Identify the Problem

- Ask the customer questions and be **respectful**.
- Use both **open-ended** and **closed-ended** questions.
  - **What are they?**
- ID Customer Information
  - Name
  - Address
  - Phone Number
- ID Computer Configuration
  - Model
  - OS
  - Network
  - Connections

"I checked the serial number of your laptop. It's a waffle iron."

Identify the Problem

1

- Listen to **beep codes**
- Use BIOS or UEFI to identify POST problems
- Tools to use:
  - LEDs
  - Error Messages
  - Event Viewer
  - Device Manager
  - Task Manager
  - Other diagnostics tools

- **LOOK, LISTEN & FEEL**

Establish a Theory of Probable Cause

2

- Create a list of the most common reasons for the error.
- List the easiest or most obvious causes at the top
- More complex causes at the bottom.
- Probable causes
  - Device powered off
  - Outlet Power switch turned off
  - Surge protector off
  - Loose cables
  - Non-bootable disk
  - Incorrect boot order

Test the Theory to Determine the Cause

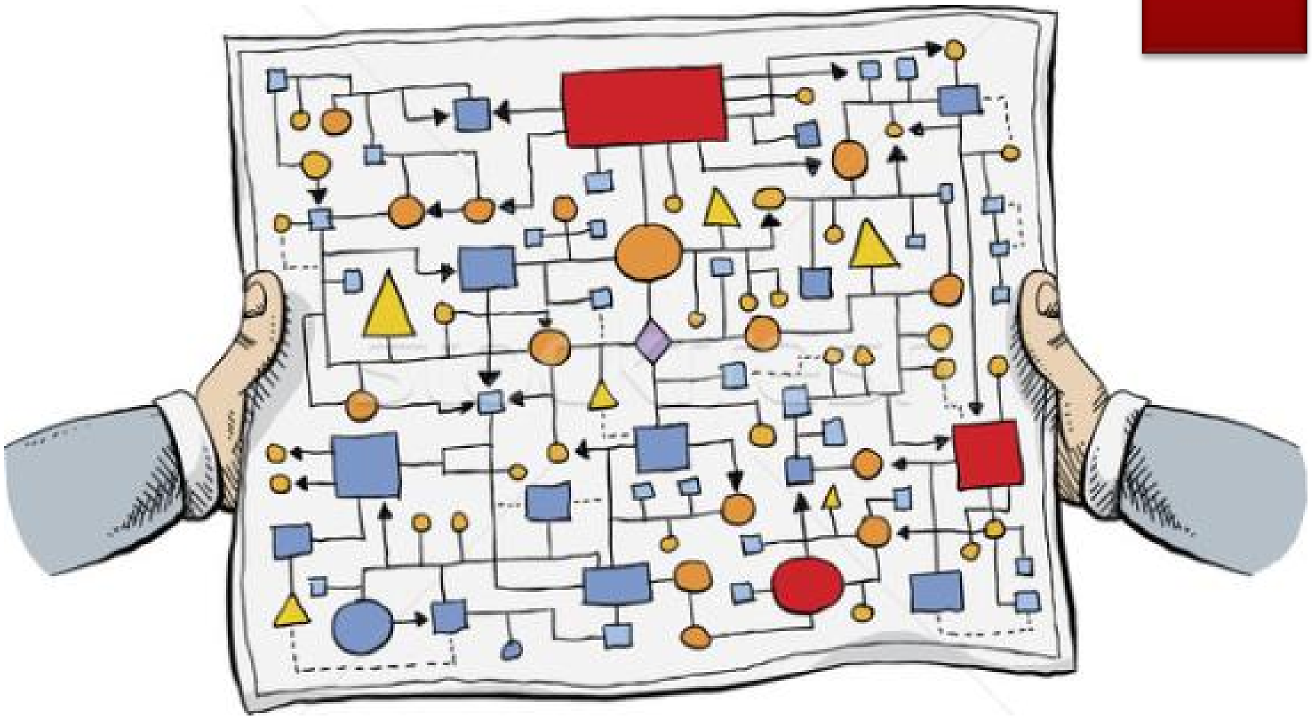
3

- Test your theories
  one at a time.
  - **Why?**
- Common Tests
  - Power On
  - Outlet Power Switch
  - Surge Protector On
  - External Cables Secure
  - Ensure Boot Drive Is Bootable
  - Verify Boot Order

Establish a Plan of Action to Resolve the Problem and Implement the Solution

4

- If no solution, more research may be needed
- **Where should you search?**
  - Helpdesk repair logs
  - Other techs
  - Manufacturer FAQs
  - Tech websites
  - News groups
  - Computer Manuals
  - Device Manuals
  - Online Forums
  - Internet Search



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Verify Full Functionality and, If Applicable, Implement Preventive Measures

- Process is not over **until full system functionality is confirmed.**
  - **What does this mean?**
- If the system is working properly, implement preventive measures if needed.
  - Reboot Computer
  - Ensure applications work
  - Verify network and Internet
  - Print a document
  - Check attached devices
  - Ensure no error messages

Document Findings, Actions, and Outcomes

6

- Discuss problem with customer, both verbally and in writing.
- Have customer verify problem is solved.
- Document the entire process for future reference.
  - Problem description
  - Steps to resolve the problem
  - Components used in the repair
  - Time spent fixing problem(s)

PC Common Problems and Solutions

- Some common hardware problems:

Video Cable Failure

USB KB & mouse switched

RAM improperly installed

CPU Fan improperly installed → case shut

Front Panel not installed

SATA data cable not installed

Power off
- Power strip
- Power supply

— power cable disconnected

Audio – disconnected
– speakers off

DVI – connected into motherboard no video card

Optical Drive
– power not connected

PC Common Problems and Solutions

- **Storage Device** - Storage device problems are often related to loose, or incorrect cable connections, incorrect drive and media formats, and incorrect jumper and BIOS settings.

PC Common Problems and Solutions

- **Motherboard and Internal Components** - These problems are often caused by incorrect or loose cables, failed components, incorrect drivers, and corrupted updates.

PC Common Problems and Solutions

- **Power Supply** - Power problems are often caused by a faulty power supply, loose connections, and inadequate wattage.

PC Common Problems and Solutions

- **CPU and Memory** - Processor and memory problems are often caused by faulty installations, incorrect BIOS settings, inadequate cooling and ventilation, and compatibility issues.

PC Common Problems and Solutions

- **Displays** – Display problems are often caused by incorrect settings, loose connections, and incorrect or corrupted drivers.

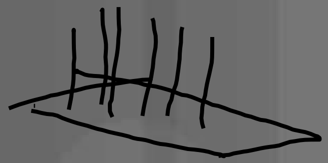
Common Problems & Solutions

- Computer problems can be attributed to hardware, software, networks, or some combination of the three.
- Common PC hardware problems include:
  - Storage device problems
  - Motherboard and internal components problems
  - Power supply problems
  - CPU and memory problems
- **Can you think of others?**

Common Problems and Solutions for Storage Devices

Select each problem

- The computer does not recognize a storage device.
- The computer does not recognize an optical disc.
- The computer will not eject the optical disc.
- The computer does not recognize a removable, external drive.
- A media reader cannot read a memory card that works properly.
- Retrieving or saving data from the USB flash drive is slow.



The computer does not recognize a storage device

Probable Causes	Possible Solutions
The power cable is loose.	Secure the power cable.
The data cable is loose.	Secure the data cable.
The jumpers are set incorrectly.	Reset the jumpers.
A storage device failed	Replace the storage device.
The storage device setting in BIOS are incorrect.	Reset the storage device settings in BIOS.

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Common Problems and Solutions for Motherboards and Internal Components

Identify the Problem

The clock on the computer is no longer keeping the correct time or the BIOS settings are changing when the computer is rebooted.

After updating the BIOS firmware, the computer will not start.

The computer displays the incorrect CPU information when the computer boots.

The hard drive LED on the front of the computer does not light.

The built-in NIC has stopped working.

The computer does not display any video after installing a new PCIe video card.

The new sound card does not work.

The clock on the computer is no longer keeping the correct time or the BIOS settings are changing when the computer is rebooted

Probable Causes	Possible Solutions
The CMOS battery may be loose.	Secure the battery.
The CMOS battery may be drained.	Replace the battery.

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Common Problems and Solutions for Power Supplies

Identify the Problem

The computer will not turn on.

The computer reboots, turns off unexpectedly; or there is smoke or the smell of burning electronics.

The computer will not turn on

Probable Causes	Possible Solutions
The computer is not plugged in to the AC outlet.	Plug the computer into a known good AC outlet.
The AC outlet is faulty.	Plug the computer into a known good AC outlet.
The power cord is faulty.	Use a known good power cord.
The power supply switch is not turned on.	Turn on the power supply switch.
The power supply switch is set to the incorrect voltage.	Set the power supply switch to the correct voltage setting.
The power button is not connected correctly to the front panel connector.	Correctly orient the power button to the front case panel connector and reconnect.
The power supply has failed.	Install a known good power supply.

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Common Problems and Solutions for CPUs and Memory

Identify the Problem

- The computer will not boot or it locks up.
- The CPU fan is making an unusual noise.
- The computer reboots without warning, locks up, or displays error messages.
- After upgrading from a single core CPU to a dual core CPU, the computer runs more slowly and only shows one CPU graph in the Task Manager.
- A CPU will not install onto the motherboard.
- The computer does not recognize the RAM that was added.
- After upgrading Windows, the computer runs very slowly.

The computer will not boot or it locks up

Probable Causes	Possible Solutions
The CPU has overheated.	Reinstall the CPU. • Replace the CPU fan. • Add fan(s) to the case.
The CPU fan is failing.	Replace the CPU fan.
The CPU has failed.	Replace the CPU.

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Common Problems and Solutions for Displays

Identify the Problem

Display has power but no image on the screen.

The images on the display have distorted geometry.

The display is flickering.

The display has a "ghost" image.

The image on the display looks dim.

Images on a display screen are distorted.

Pixels on the screen are dead, or not generating color.

The monitor has oversized images and icons.

The image on the screen appears to flash lines or patterns of different color and size (artifacts).

In a multiple monitor setup, the displays are not aligned or are incorrectly oriented.

Color patterns on a screen are incorrect.

The projector overheats and shuts down.

The display is in VGA mode.

Display has power but no image on the screen

Probable Causes	Possible Solutions
Video cable is loose or damaged.	Reconnect or replace video cable.
The computer is not sending a video signal to the external display.	Use the Fn key along with the multi-purpose key to toggle to the external display.

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Personal Reference Tools

- Personal reference tools include troubleshooting guides, manufacturer manuals, quick reference guides, and repair journals. In addition to an invoice, a technician keeps a journal of upgrades and repairs:
- **Notes** - Make notes as you go through the troubleshooting and repair process. Refer to these notes to avoid repeating steps and to determine what needs to be done next.
- **Journal** - Include descriptions of the problem, possible solutions that have been tried to correct the problem, and the steps taken to repair the problem. Note any configuration changes made to the equipment and any replacement parts used in the repair. Your journal, along with your notes, can be valuable when you encounter similar situations in the future.
- **History of repairs** - Make a detailed list of problems and repairs, including the date, replacement parts, and customer information. The history allows a technician to determine what work has been performed on a specific computer in the past.

Advanced Problems and Solutions for Hardware

Instructions

Click a problem on the left side of the screen to see probable causes and possible solutions. At any time, click another problem on the left side of the screen. To see a PDF of the entire table, click the Show PDF button on the lower right corner of the screen.

Identify the Problem

- RAID cannot be found.
- RAID stops working.
- The computer exhibits slow performance.
- The computer does not recognize a removable external drive.
- After updating the CMOS firmware, the computer will not start.
- The computer reboots without warning, locks up, or displays error messages or the BSOD.
- After upgrading from a single core CPU to a multi-core CPU, the computer runs slower and only shows one CPU graph in Task Manager.

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Lab – Using a Multimeter and a Power Supply Tester

In this lab, you will learn how to use and handle a multimeter and a power supply tester.

Lab – Troubleshoot Hardware Problems

In this lab, you will diagnose the cause of various hardware problems and solve them.

4.3 Chapter Summary

Conclusion

Chapter 4: Preventive Maintenance and Troubleshooting

4.1 Preventive Maintenance

- Explain why preventive maintenance must be performed on personal computers.
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4.2 Troubleshooting Process

Troubleshoot problems with PC and Peripheral devices

- Describe each step of the troubleshooting process.
- Identify common problems and solutions for PCs.
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